

Learning Dynamics for High-Speed Tractor-Trailer Control

Aaron Chen* · David Shen* · Taekyung Kim* · Kaleb Ben Naveed · Dimitra Panagou
University of Michigan, Ann Arbor, USA | EECS · Robotics · Aerospace Engineering

VIDEO / CODE
QR placeholder

Introduction & Motivation

Model-based control depends on accurate dynamics.

Opportunity — Learn what analytic models miss

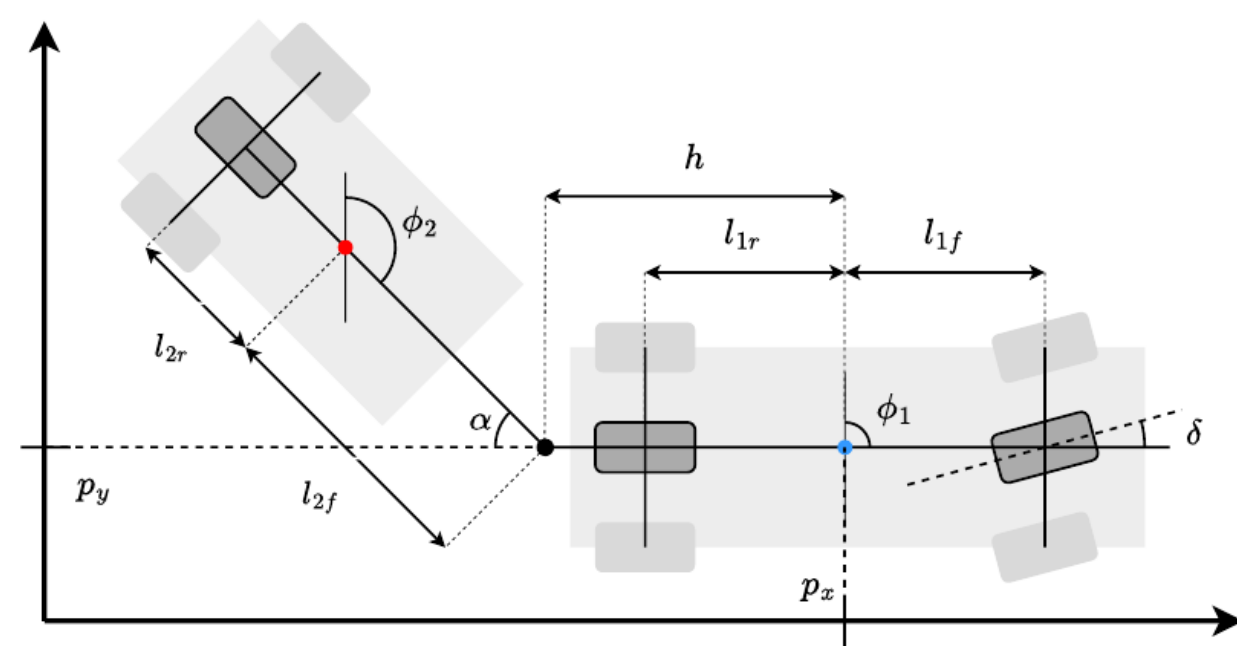
Learned dynamics can improve prediction when system properties and environmental conditions are uncertain.

Testbed — Coupled, unstable vehicle dynamics

Tractor-trailers exhibit nonlinear dynamics and sensitivity to friction and terrain, with instability especially pronounced in reverse.

Gap — High-speed reverse control under uncertainty

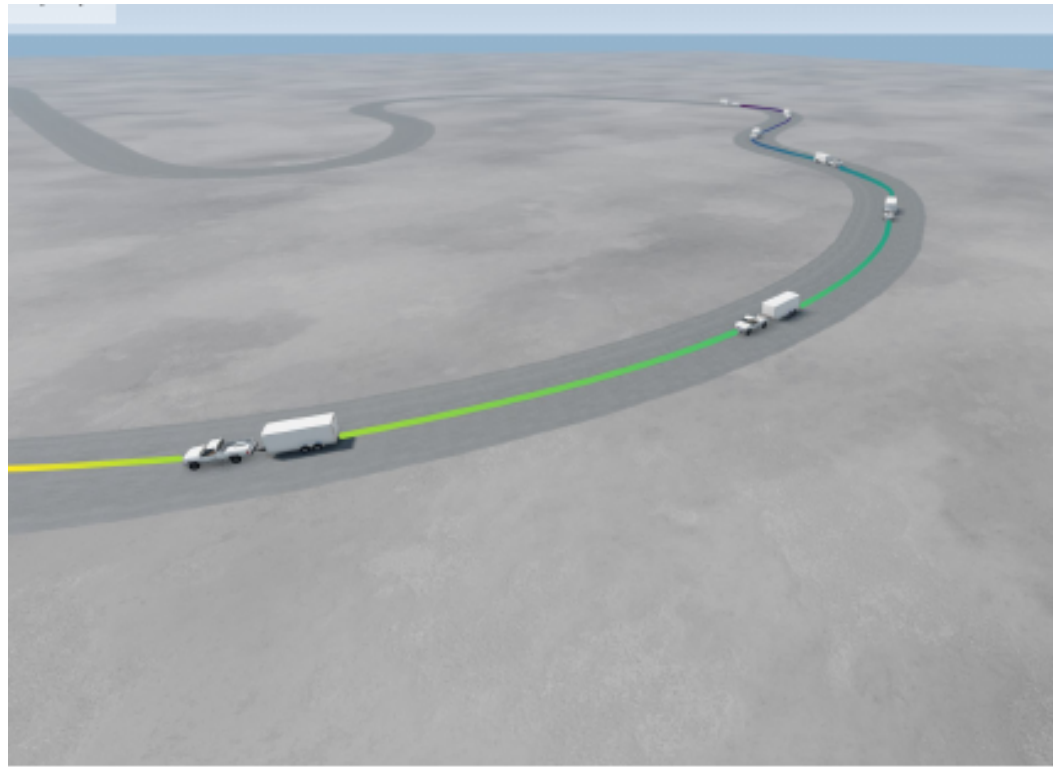
Learned predictive control remains underexplored in this demanding operating regime.



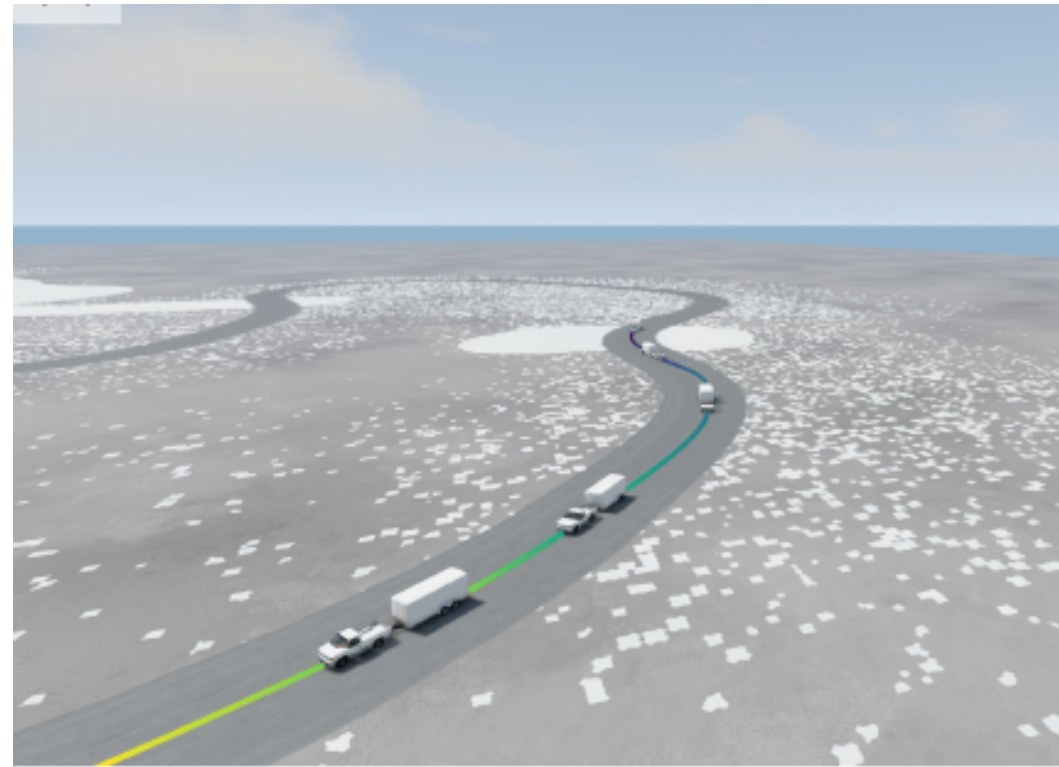
A demanding testbed for learning dynamics under uncertainty.

Experimental Setup

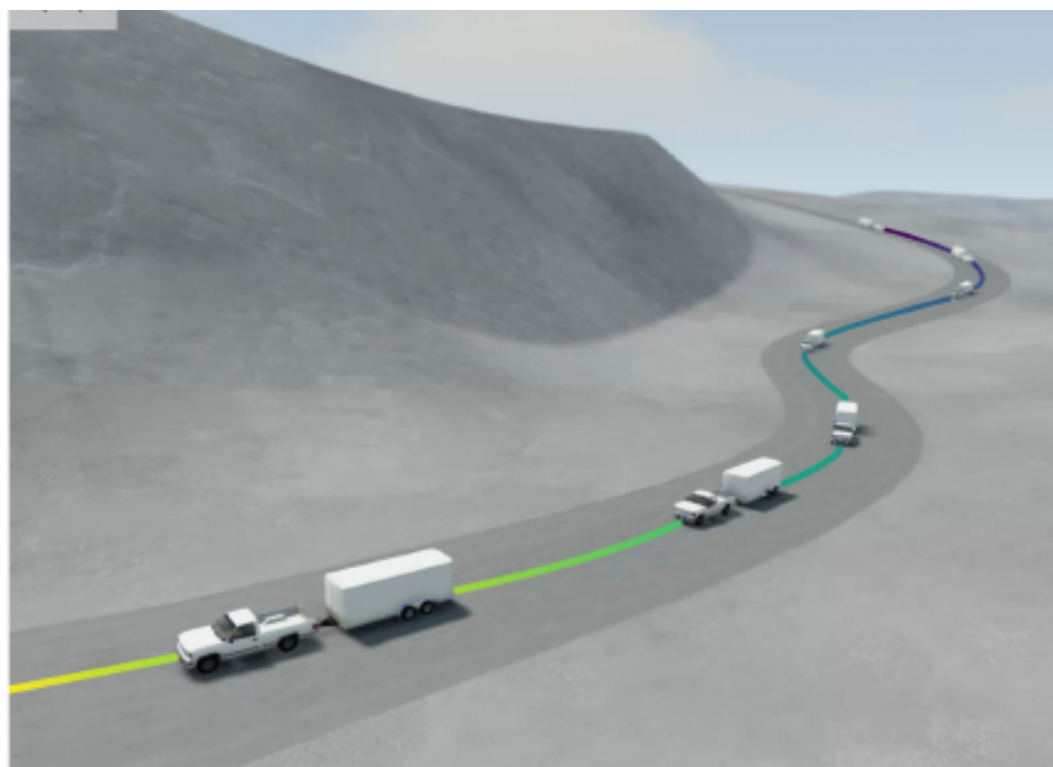
(a) Asphalt



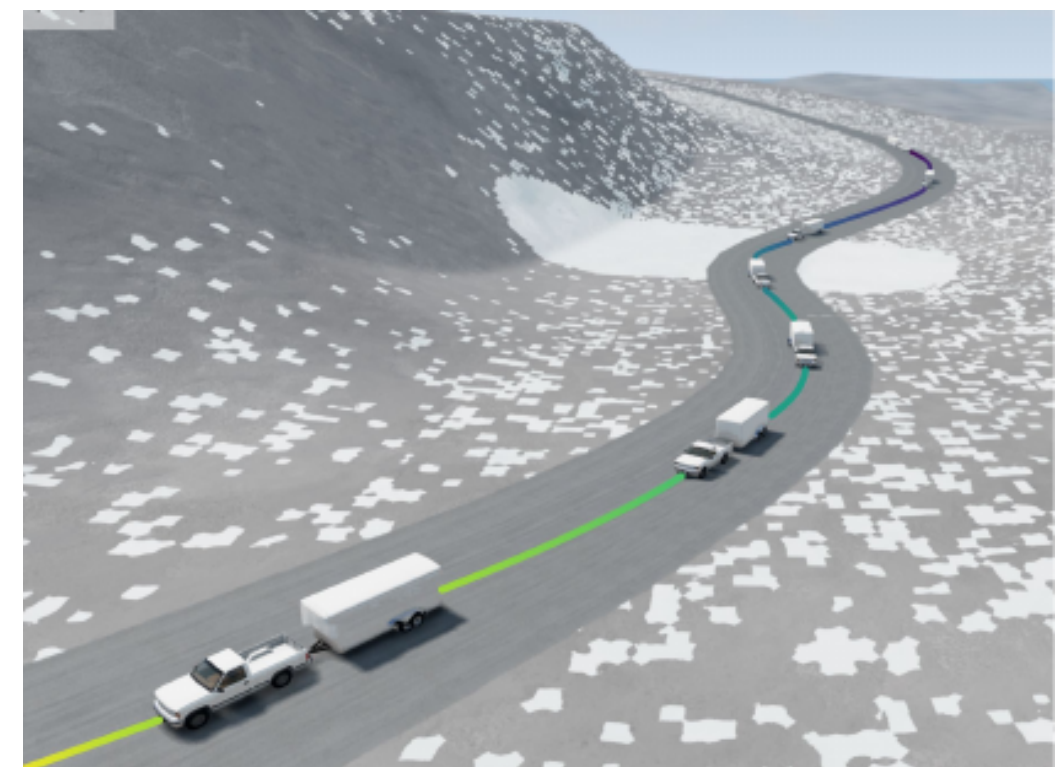
(b) Snow



(c) Slope



(d) Snow + Slope

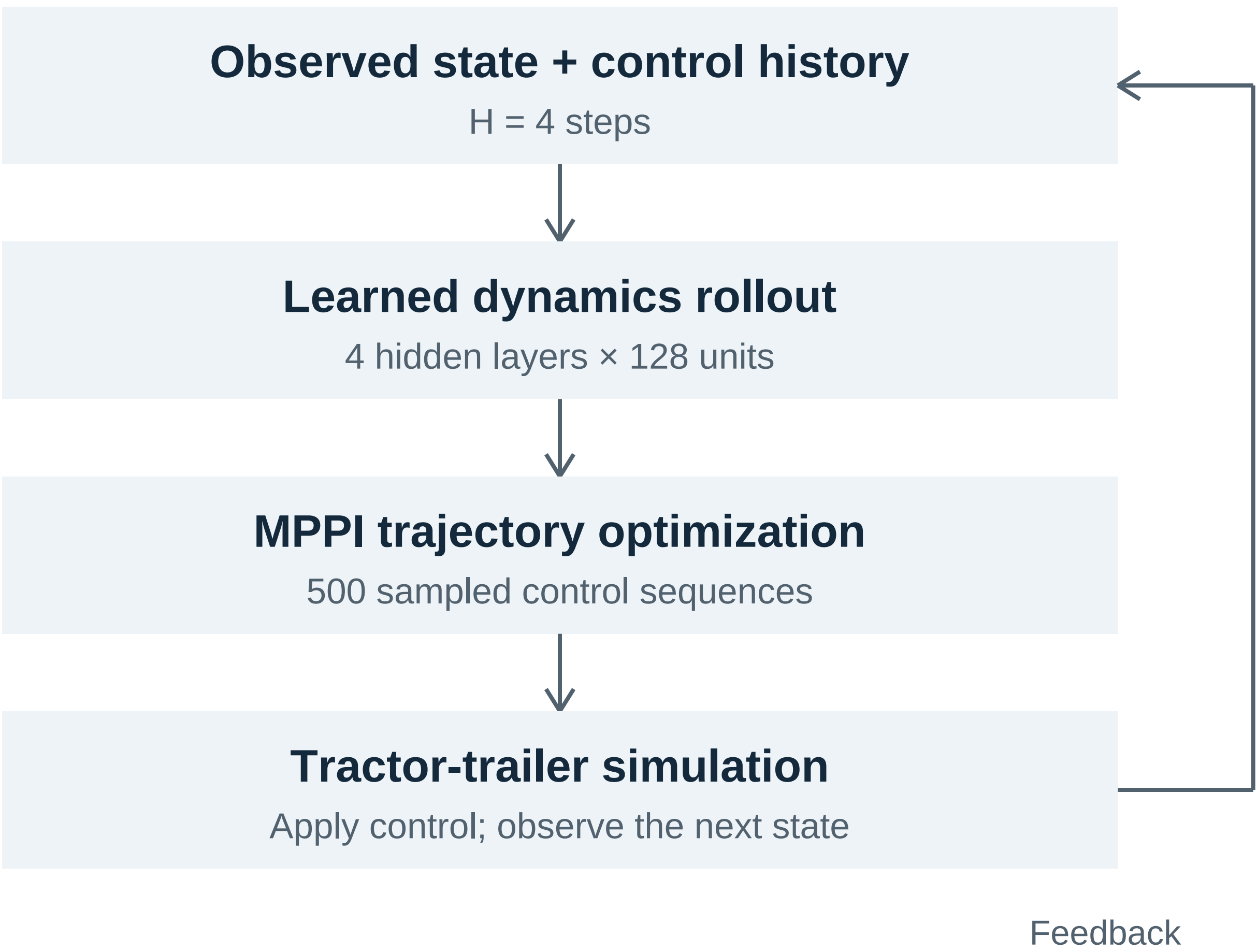


Representative reverse-driving trajectories with learned-dynamics MPPI in BeamNG.tech.

Targets: -30 / -50 km/h reverse; $+80$ km/h forward.
Trials: 9 runs per configuration, up to 500 steps.
Baselines: Fiala-MPPI, LQR-PID, IPOPT-MPC, MPCC.

Learning-Based MPPI

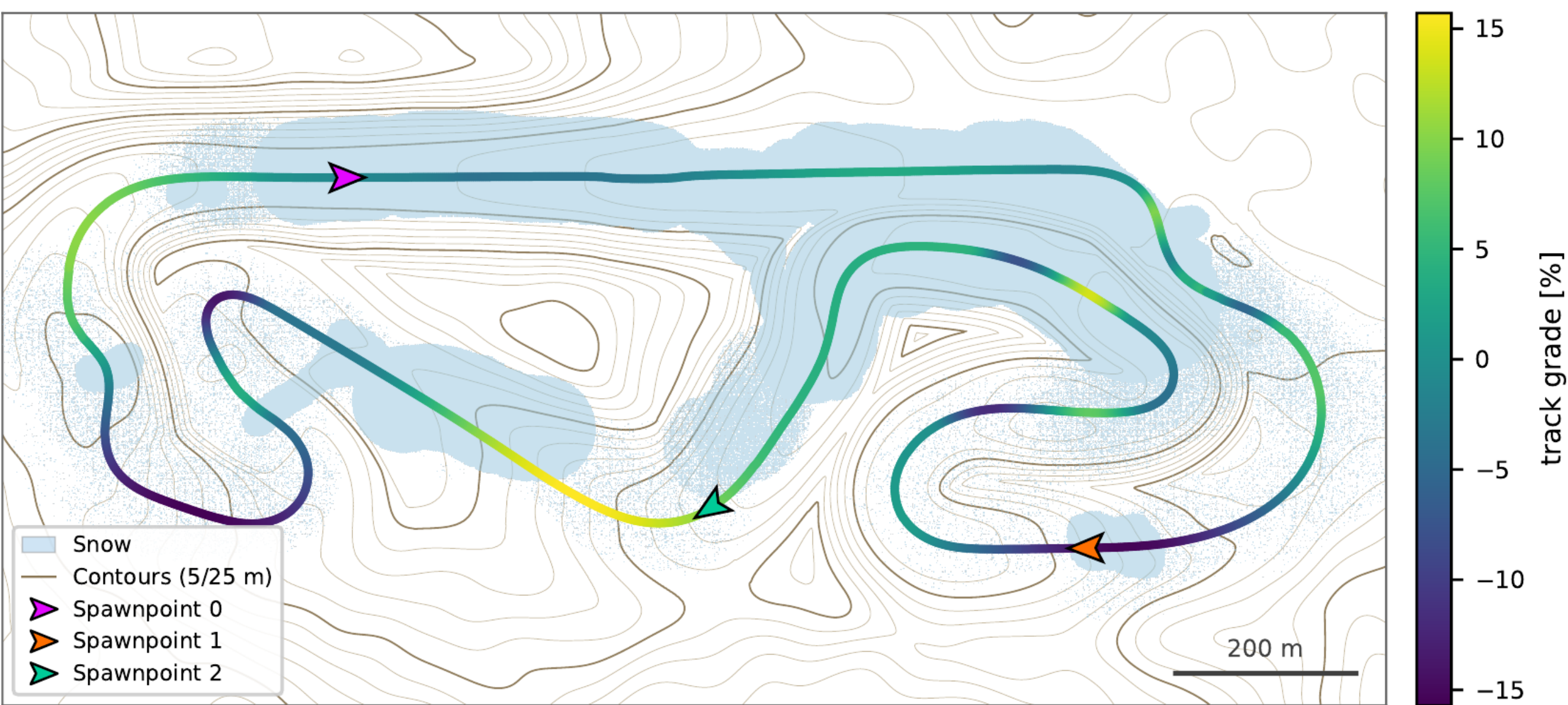
An MLP predicts vehicle dynamics from recent observations and controls. MPPI uses these predictions to evaluate candidate control sequences.



Objective: track target speed and the road centerline while penalizing hitch angle and constraint violations.

Training & Terrain Transfer

- Collect trajectories across friction levels and target speeds; add data near handling limits.
- Predict accelerations and realized actuator states, then integrate to obtain rollout states.
- Slope and banking are absent from training.



Closed-Loop Results

Learned MPPI improves mean-speed tracking and RMS hitch angle over Fiala-MPPI in all 12 environment-velocity configurations.

Completed runs by environment

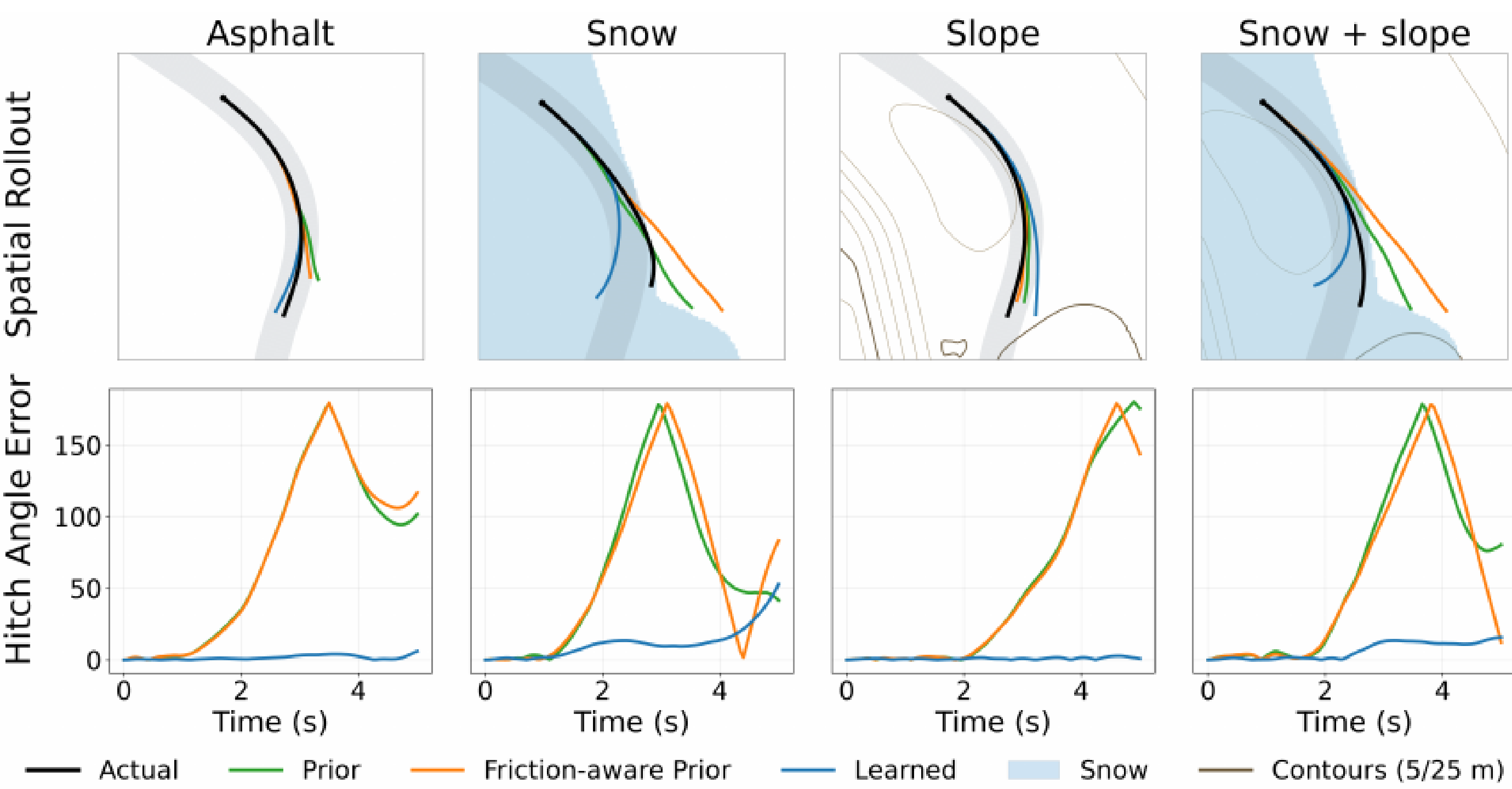
Three velocity setpoints pooled · maximum 27 runs per cell

Controller	Asphalt	Snow	Slope	S + S
Learned MPPI	27	27	25	19
Fiala-MPPI	20	19	23	15
LQR-PID	20	18	21	15
IPOPT-MPC	14	11	16	13
MPCC	4	4	5	4

S + S: Snow + Slope. Counts include every run.

54/54 runs complete on flat Asphalt and Snow. On Snow + Slope, reverse-driving stalls limit performance.

Multi-Step Prediction



Forward-driving rollout examples from the paper: trajectories (top) and hitch-angle error over time (bottom).

- Lowest position RMSE in 4/4 reverse environments.
- Lowest hitch-angle RMSE in 4/4 forward environments.
- Lowest reported RMSE in 13/16 comparisons overall.